Synergistic Role of RGG2 in Modulating FUS RRM Structure and RNA Binding Revealed by Enhanced Sampling Simulations

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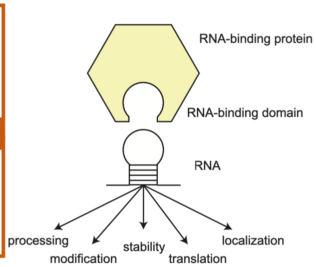
Background and Motivation

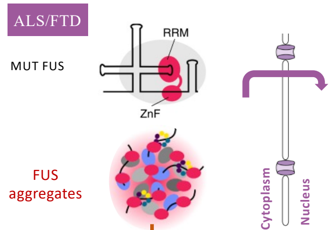
Fused in sarcoma (FUS) as an example of RNA-binding proteins

FUS plays a key role in **RNA splicing**, especially through interactions with U1 small nuclear (sn) RNA.

FUS aggregation is a pathological hallmark of Amyotrophic Lateral Sclerosis (ALS) and Frontotemporal Dementia (FTD)



Hentze, Matthias W., et al. Nature reviews Molecular cell biology 19.5 (2018): 327-341.



>> Impaired snRNP assembly

>> Downstream defects in RNA processing

>> Motor neuron degeneration

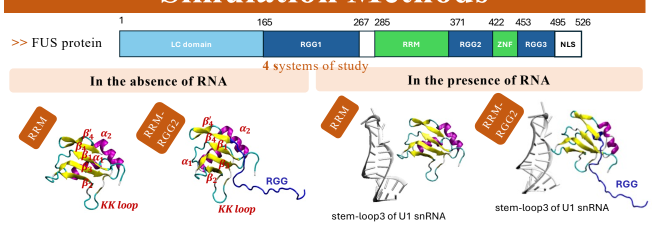
Jutzi, Daniel, et al. Nature Communications 11.1 (2020): 6341.

The **folding stability** of the FUS RNA-recognition motif (RRM) is biophysically critical, as its **irreversible unfolding** triggers aggregation.

The RRM domain is significantly perturbed by the presence of its neighboring domains and RNA

What are effects of **RGG and RNA** on RRM stability?

Simulation Methods



>> Initial structures: Taken from PDB ID 6SNJ

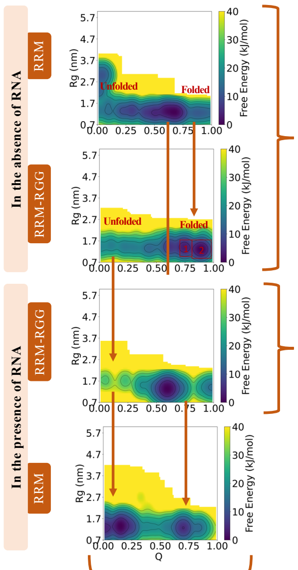
>> CHARMM36 and TIP3P modified water force fields

>> Sampling method: MM-OPES (Multithermal-Multiumbrella On-the-fly Probability Enhanced Sampling)

>> Order parameters (Q: fraction of native-like contacts and distance between centers of masses for RNA-protein systems)

Results and Discussions

Structural changes of RRM in the presence of RGG, RNA, or both

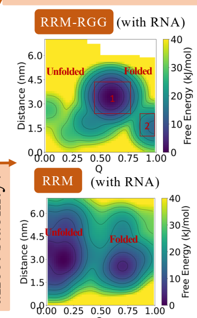


>> **Higher stability** for the folded RRM in the presence of RGG
>> **Same compactness** between folded state of RRM and RRM-RGG

Does RGG help secondary structure formation? (Yes!)

>> No stable basin for Q < 0.25
>> Q ≈ 0.6 is again the most stable basin with the RNA, similar to RRM alone case!

How does RGG mediate RNA interactions?

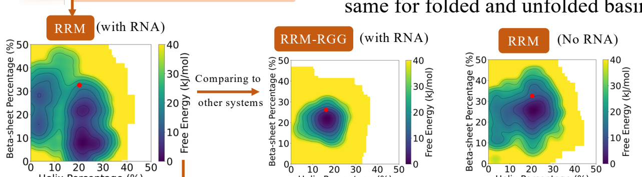


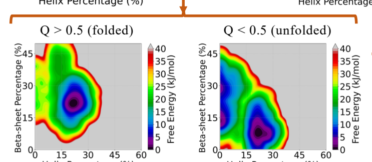
Does RNA proximity affect stability?

>> RNA-RRM distance stays the same for folded and unfolded basins.

>> The folded basin Q slightly exceeds the RRM-RGG system's but matches the RNA-free RRM
>> The emergence of very stable basin for Q < 0.25.

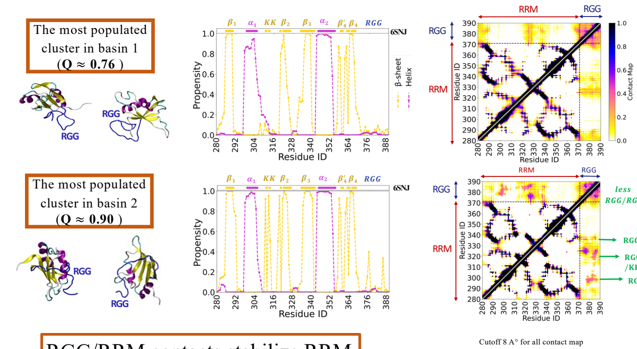
How do helices and beta sheets contribute?





RNA/RRM interactions make the unfolded state more structured (but heterogeneous) when RGG is absent.

>> Analyzing two folded basins of RRM-RGG in the **absence** of RNA

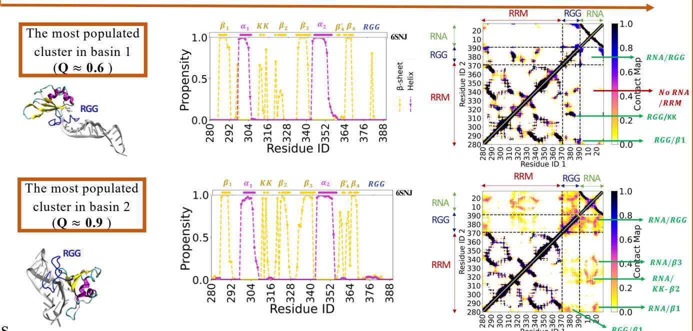


RGG/RRM contacts stabilize RRM secondary structures.

RGG acts as a stabilizing belt for RRM in Q ≈ 0.90

Rahimi, Kosar, et al. The Journal of Physical Chemistry B 130.27 (2026): 6773-6781.

>> Analyzing two folded basins of RRM-RGG in the **presence** of RNA



In the most stable basin (Q ≈ 0.60), RNA is freezing RGG with the terminal nucleotides and has has almost no interactions with RRM.

RNA/RGG interacts with RRM in the Q ≈ 0.9 basin.

Conclusions

>> The interactions of disordered region (RGG) with an adjacent globular domain (RRM) enhances the folding stability.
>> RNA shows more stable interactions with RGG than RRM when both present, mostly through its terminal nucleotides.
>> The absence of the RGG changes the way RNA interacts with RRM, making RRM to sample more structured unfolded configurations.

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